Final Project Report

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Hotel Price Analysis in Yerevan: Identifying the Key Drivers of Hotel Pricing

Introduction

The hotel industry in Yerevan has expanded rapidly in the last decade because of increasing tourism. As competition intensifies, hotels must understand how their pricing compares to the market and which factors influence customers' willingness to pay. A data-driven approach to pricing can help hotels make more effective decisions, improve revenue, and create a clearer value proposition for their guests.

Purpose

This project analyzes a dataset of hotels in Yerevan containing information on star rating, customer satisfaction scores, service types (e.g. free parking, airport shuttle, spa, and fitness center), and the daily room price for two adults. The main question is:

"Which factors most strongly influence hotel prices in Yerevan?"

By identifying the drivers of hotel prices, the project provides actionable insights that can help hotels improve their pricing strategy, offer the right services, improve guest satisfaction, and strengthen their competitive advantage in the Yerevan hospitality market.

Business Strategy

This project supports a pricing optimization strategy for hotels in Yerevan. By identifying the attributes that most strongly influence hotel prices, hotels can assess whether they are underpriced or overpriced relative to competitors and adjust pricing accordingly. This enables better market positioning, improved revenue management, and more informed investment decisions in high-impact amenities. Overall, the strategy helps hotels allocate resources more efficiently while strengthening their competitive advantage.

Data Description

I will use the "Yerevan Hotel Reservation Prices" dataset from Kaggle (link here.)

The dataset includes information like hotel names and star categories, customer satisfaction ratings (rating out of 10 for star, location, cleanliness, etc.), overall review score, amenities (free parking, fitness, spa, etc.), daily room price for two adults.

I have also applied data cleaning steps such as handling missing values, converting "Yes/No" fields into binary variables, and detecting outliers in pricing.

Model Selection

I have used a few analytical methods from this course:

1. **Descriptive Analytics and Visualization in Tableau**

   I have visualized the price distribution across hotels, average price by star rating, correlation heatmap of rating vs. price, and top 10 most expensive hotels and their ratings. These descriptive visuals help identify patterns in the pricing structure.

2. **Regression Analysis**

   I have built a multiple linear regression model using Python to analyze which features significantly affect price, such as Star Rating, Overall Rating, Amenity Indicators (like airport shuttle or fitness centre).

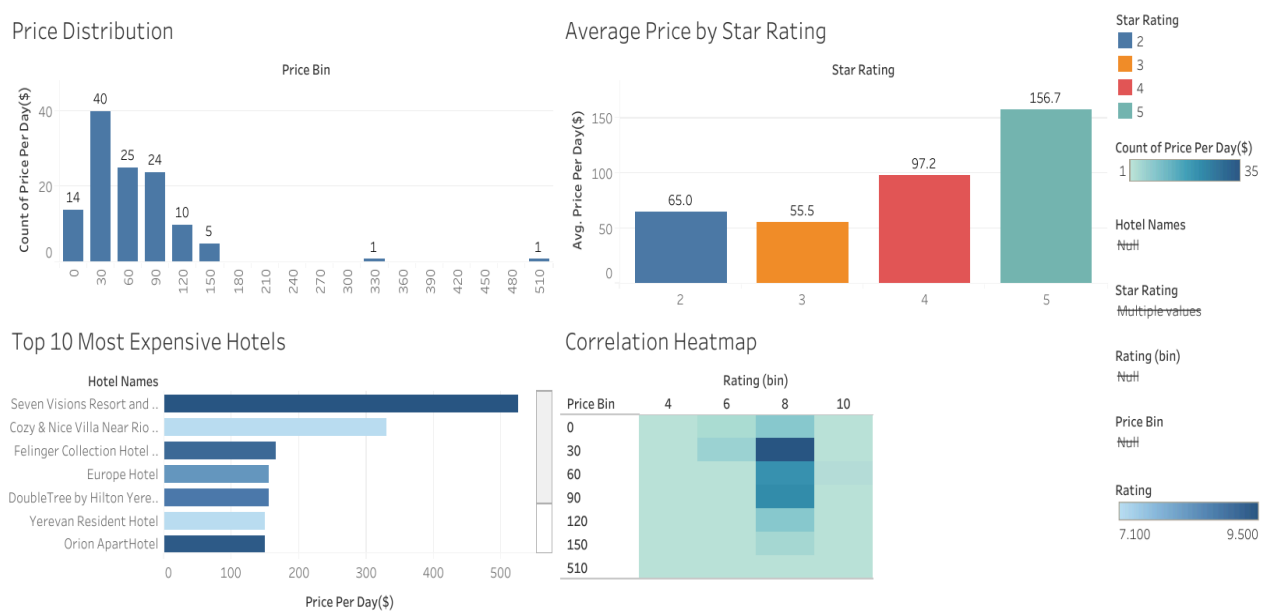
3. **Clustering**

   I have applied K-means clustering to group hotels into segments. Clusters can support segmentation-based pricing recommendations.

Results and Analysis

Descriptive Analysis and Visualization in Tableau

To explore the structure of hotel prices in Yerevan and identify initial patterns, I created several descriptive visualizations in Tableau. These charts provide an initial understanding of how prices vary across hotels, star categories, rating and amenities before applying statistical models.



Summary of Tableau Insights

The Tableau dashboard reveals several important descriptive patterns. Most hotels fall into the affordable or mid-range categories, with fewer extremely high-priced hotels. Star rating is a

strong indicator of price, particularly between 4 star and 5 star hotels. Guest ratings correlate with price, but not strongly enough to be the sole determinant. The premium market is small but significantly more expensive.

Regression Results

To analyze how hotel characteristics influence pricing, I built a multiple linear regression model using daily room price as the dependent variable. The scikit-learn model achieved an R^2 of 0.3068 and an MSE of 1134.79, indicating that approximately 30.7% of price variation is explained by the selected features. Given the absence of external factors such as seasonality and brand effects, this level of explanatory power is expected. To assess statistical significance, I also ran the model using Statsmodels, producing a higher R^2 of 0.481 and an adjusted R^2 of 0.427, suggesting a reasonably good fit to the full dataset.

Significant Predictors

Based on p-values at the 5% significance level ($p < 0.05$), several predictors are statistically significant. Overall rating, a rather unusual find, shows a negative relationship with price, suggesting that lower-priced hotels may compete through higher guest satisfaction, while some higher-priced hotels receive more moderate ratings. In contrast, the presence of a fitness centre is associated with a substantial price increase of approximately $47.7 per night, making it one of the strongest positive predictors. Airport shuttle services and higher location scores also have positive and significant effects on price. Comfort score has the largest impact, with an estimated increase of approximately $80 per point, highlighting the importance of room quality in pricing decisions.

Non-Significant Predictors

Variables such as Free Parking, Spa and Wellness, Staff, Facilities, and Cleanliness were not statistically significant. These features may still influence customer satisfaction but do not directly translate to higher prices in a measurable way.

Key Insight

Overall, the regression results show that comfort, fitness facilities, location, and shuttle services are the most important contributors to hotel pricing in Yerevan. Meanwhile, customer ratings appear to behave differently due to potential multicollinearity or compensatory behavior in lower-priced hotels.

Regularized Regression

To address multicollinearity and improve model stability, I applied Lasso and Ridge regression. These models introduce penalties that shrink irrelevant or unstable coefficients.

The results were:

- Lasso R^2: 0.3624
- Ridge R^2: 0.3432

Both regularized models outperformed the baseline model (scikit-learn model), with Lasso performing the best. This indicates a few things such as, some predictors contain overlapping information, Lasso's feature selection effect improves generalization, and regularization helps stabilize coefficients in models with correlated rating variables

These findings reinforce that while several hotel characteristics contribute to pricing, only a subset improves predictive performance.

Clustering Results

To identify natural market segments within Yerevan hotel landscape, I applied K-means clustering using the same set of hotel attributes. These segments were the result:

Cluster	Average Price (USD)
Cluster 0	67.03
Cluster 1	72.34
Cluster 2	110.00

Cluster Interpretation

- Cluster 0 are the Budget Hotels (~$67/day). Typically lower-star hotels with fewer amenities and moderate customer review scores. This group represents the budget segment.
- Cluster 1 are the Mid-Range Hotels (~$72/day). These hotels show slightly higher satisfaction ratings and better service quality, offering improved value without high pricing.
- Cluster 2 are Premium Hotels (~$110/day). This segment includes hotels with strong comfort scores, better amenities, and likely higher star scores.

Segmentation Insight

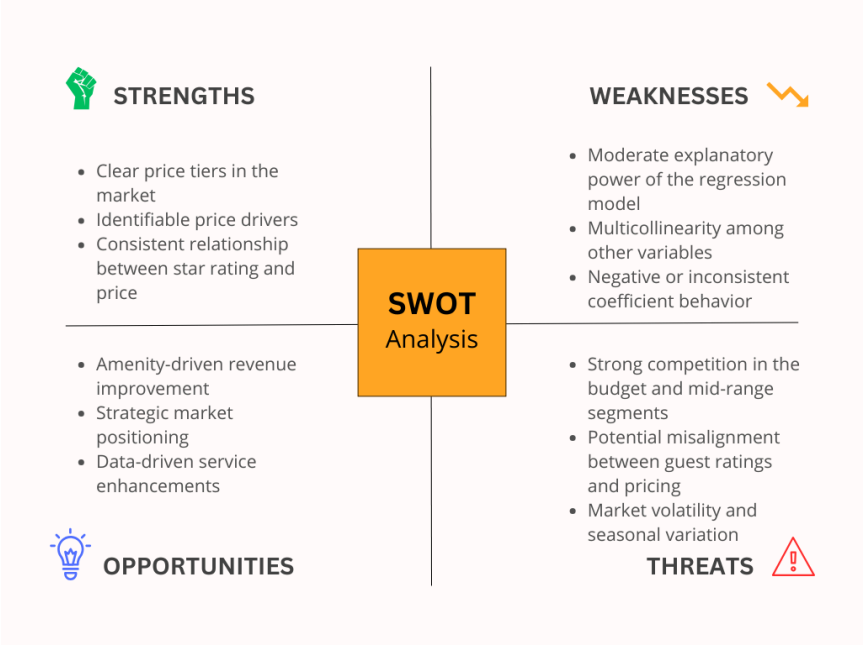
The clusters highlight three distinct market tiers in Yerevan. Hotels can use this segmentation to position themselves against similar competitors and identify where strategically adding amenities or improving comfort levels may justify higher pricing.

Summary of Findings

Across all analytical methods, the results indicate a few noteworthy findings. First, comfort location scores, and fitness centres have the largest positive impact on hotel prices. Also, overall rating and cleanliness indicators behave differently, possibly due to multicollinearity. Lasso regression improves model performance, indicating that some predictors provide limited value. Finally, K-means clustering reveals three meaningful hotel segments, supporting segmentation-based pricing strategies. These findings provide valuable insights for hotel managers seeking to refine pricing strategies and understand competitive positioning in Yerevan's hospitality market.

SWOT Analysis

A SWOT analysis was conducted to evaluate the strategic implication of the findings for hotels operating in Yerevan. The strengths, weaknesses, opportunities, and threats below reflect insight from descriptive analysis, regression models, and clustering results.



Decision and Financial Analysis

To complement the analytical results, I evaluate one actionable business decision using financial reasoning. Specifically, I examine whether investing in a new hotel amenity could be financially justified based on predicted impact on pricing. This approach connects the regression findings with practical decision making and profitability analysis.

Scenario: Adding a Fitness Centre

The regression analysis showed that the presence of a fitness centre is associated with an average price increase of approximately $47.7 per night. To determine whether it is financially feasible for a hotel to invest in a new fitness facility, I evaluate the potential return using a simple payback period approach. I assume that the average occupancy rate of hotels in Yerevan is 70% (from here and here), regression's estimated price increase from fitness center is $47.7 per night, and the estimated installation cost is $30000 (from here).

Revenue Impact Calculation

Additional annual revenue: Prince increase * Occupancy rate * Days per year = 47.7 * 0.7 * 365 = $12187.35 per year

Payback Period

Payback period = Installation cost / Additional annual revenue = 30000 / 12187.35 = 2.462 years

Interpretation

A payback period of approximately 2 years and 6 months indicates that adding a fitness centre is a financially attractive investment. Hotels in the mid-range or premium segments can recover installation costs relatively quickly and benefit from higher long-term revenue. This aligns with clustering results, where premium hotels command significantly higher prices and typically possess more amenities. Adding such an amenity not only increases revenue potential but may also strengthen a hotel's competitive advantage within its market segment. This is backed by the regression results as well.

Management Implication

Based on the financial evaluation, hotels lacking wellness facilities can consider adding a fitness centre as a strategic investment. This amenity enhances perceived value, leads to a higher pricing power, and differentiates the hotel from competitors in the crowded budget and mid-range segments.

Recommendations and Limitations

Based on the analysis, a few recommendations can be made for hotel managers in Yerevan. Improving room comfort should be a priority, as comfort was the strongest predictor of price, with an estimated impact of approximately $80 per night and high potential return on investment. Hotels should also prioritize high-impact amenities, particularly fitness centres and airport shuttle services, which significantly increase pricing power and can potentially have relatively short payback periods as shown in the financial analysis scenario. Finally, hotels should evaluate their pricing relative to market segments identified through clustering to ensure alignment with direct competitors and capture additional revenue opportunities.

Despite providing useful insights, the analysis has several limitations. The dataset includes only 118 hotels, which may not fully capture seasonal or long-term pricing determinants. Additionally, key external factors such as room type, seasonal demand, and booking timing are not included, likely explaining part of the unexplained variation in prices. Multicollinearity among rating variables affects coefficient stability, although regularization partially fixes this issue. Finally, the regression model relies on standard linear assumptions that were not fully tested, which may influence the reliability of the estimated effects.

Conclusion

This project analyzed the key drivers of hotel pricing in Yerevan using descriptive analytics, regression models, and clustering techniques. The results show that comfort, fitness facilities, and transportation services have the strongest positive impact on prices, while amenities such as parking or spa access play a smaller role. The market clearly segments into budget, mid-range, and premium tiers with distinct pricing patterns. Despite moderate explanatory power due to missing external factors, the findings provide clear strategic guidance for hotels seeking to optimize pricing and prioritize high-impact improvements.